A NEW FORMULA FOR SUMS OF POWERS USING CENTRAL NEWTON'S INTERPOLATION

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Abstract.

Contents

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Preliminaries

Lemma 0.1 (Stirling form of Binomial)**.** *For integers n, k*

$$\binom{n}{k} = \sum_{j=0}^{k} \frac{(-1)^{k-j}}{k!} \begin{bmatrix} k \\ j \end{bmatrix} n^j$$

1. Historical context

2. Central Newton's formula

Proposition 2.1. (Central Newton's formula [**?**, p. 462]). *For arbitrary integers x, and t*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!} \delta^k f(t),$$

where $\delta^k f(t)$ denotes central finite difference

$$\delta^k f(t) = \sum_{j=0}^{k} (-1)^j \tbinom{k}{j} f(t + \tfrac{k}{2} - j).$$

Definition 2.2 (Central factorials)**.** *For integers n, k*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n\left(n + \frac{k}{2} - 1\right)\left(n + \frac{k}{2} - 2\right) \cdots \left(n - \frac{k}{2} + 1\right) = n\left(n + \frac{k}{2} - 1\right)_{k-1}, & \text{if } k > 0, \end{cases}$$

where $\left(n + \frac{k}{2} - 1\right)_{k-1} = \prod_{j=1}^{k-1}\left(n + \frac{k}{2} - j\right)$ are falling factorials.

3. Definitions and notations

Definition 3.1 (Multifold sums of powers)**.** *For non-negative integers r, n, m*

$$\Sigma^0\, n^m = n^m,$$

$$\Sigma^1\, n^m = \Sigma^0\, 1^m + \Sigma^0\, 2^m + \cdots + \Sigma^0\, n^m,$$

$$\Sigma^{r+1}\, n^m = \Sigma^r\, 1^m + \Sigma^r\, 2^m + \cdots + \Sigma^r\, n^m.$$

4. Helper lemmas

Lemma 4.1 (Binomial form of central factorials)**.** *For integers n and $k \geq 1$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1}.$$

Proof. We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!}\left(n+\frac{k}{2}-1\right)_{k-1} = \frac{n}{k(k-1)!}\left(n+\frac{k}{2}-1\right)_{k-1} = \frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1},$$

because of the identity in falling factorials $\frac{(x)_n}{n!} = \binom{x}{n}$. □

Lemma 4.2 (Halved central factorials)**.** *For integers $n, k \geq 0$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} = \frac{1}{2}\binom{n+\frac{k}{2}}{k} + \frac{1}{2}\binom{n+\frac{k}{2}-1}{k}.$$

Proof. We have $\binom{a}{k} = \frac{a}{k}\binom{a-1}{k-1}$. Let $a = n + \frac{k}{2}$, then

$$\binom{n+\frac{k}{2}}{k} = \frac{n+\frac{k}{2}}{k}\binom{n+\frac{k}{2}-1}{k-1} = \frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} + \frac{1}{2}\binom{n+\frac{k}{2}-1}{k-1}.$$

Thus,

$$\frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} = \binom{n+\frac{k}{2}}{k} - \frac{1}{2}\binom{n+\frac{k}{2}-1}{k-1}.$$

By moving under common denominator, and applying binomial recurrence yields

$$\frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} = \frac{1}{2}\left[2\binom{n+\frac{k}{2}}{k} - \binom{n+\frac{k}{2}-1}{k-1}\right] = \frac{1}{2}\left[\binom{n+\frac{k}{2}}{k} + \binom{n+\frac{k}{2}-1}{k} + \binom{n+\frac{k}{2}-1}{k-1} - \binom{n+\frac{k}{2}-1}{k-1}\right].$$

Thus, the statement follows. □

Proposition 4.3 (Central Newton's formula for powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$n^m = \sum_{k=0}^{m} \frac{(n-t)^{[k]}}{k!}\delta^k t^m.$$

Proposition 4.4 (Halved central factorial powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}}{k} + \binom{n-t+\frac{k}{2}-1}{k}\right]\delta^k t^m$$

Proposition 4.5 (Generalized hockey-stick identity)**.** *For integers a, b and j,*

$$\sum_{k=a}^{b}\binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have, $\sum_{k=a}^{b}\binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^{b}\binom{k}{j}\right) - \left(\sum_{k=0}^{a-1}\binom{k}{j}\right)$. By hockey stick identity $\sum_{k=0}^{n}\binom{k}{j} = \binom{n+1}{j+1}$ yields,

$$\sum_{k=a}^{b}\binom{k}{j} = \left(\sum_{k=0}^{b}\binom{k}{j}\right) - \left(\sum_{k=0}^{a-1}\binom{k}{j}\right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. □

Lemma 4.6 (Centered hockey-stick identity)**.** *For integers $n \geq 1$, and arbitrary integers t, k, r*

$$\sum_{j=1}^{n}\binom{j-t+\frac{k}{2}+r}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Proof. By setting $j = 1-t+\frac{k}{2}+r$, and $b = n-t+\frac{k}{2}+r$ yields

$$\sum_{j=1}^{n}\binom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r}\binom{j}{k}.$$

By Generalized hockey-stick identity (4.5),

$$\sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r}\binom{j}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Thus, the claim follows. □

5. Ordinary sums of powers

Proposition 5.1 (Ordinary sums of powers). *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^1 n^m = \frac{1}{2}\sum_{k=0}^{m} \delta^k t^m \sum_{j=1}^{n} \left[\binom{j-t+\frac{k}{2}}{k} + \binom{j-t+\frac{k}{2}-1}{k}\right]$$

Proposition 5.2 (Ordinary sums of powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^1 n^m = \frac{1}{2}\sum_{k=0}^{m} \left[\binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1}\right]\delta^k t^m$$

Proposition 5.3 (Central factorial numbers sums). *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^1 n^m = \frac{1}{2}\sum_{k=0}^{m} \left[\binom{n+\frac{k}{2}+1}{k+1} - \binom{1+\frac{k}{2}}{k+1} + \binom{n+\frac{k}{2}}{k+1} - \binom{\frac{k}{2}}{k+1}\right] k! T(m,k),$$

where $T(n,k) = \frac{\delta^k 0^m}{k!}$ denote Central factorial numbers of the second kind (in Riordan's notation, see [?, p. 213]*, and* [?]*.), defined by central Newton's formula for powers, evaluated in $t = 0$*

$$n^m = \sum_{k=0}^{\infty} T(m,k) n^{[k]}.$$

Example 5.4 (Sums of squares in central differences).

$$t=0: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(1+n) + \frac{\delta^2 t^2}{2}\frac{n(1+3n+2n^2)}{6},$$

$$t=1: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-1) + \frac{\delta^2 t^2}{2}\frac{n(1-3n+2n^2)}{6},$$

$$t=2: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-3) + \frac{\delta^2 t^2}{2}\frac{n(13-9n+2n^2)}{6},$$

$$t=3: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-5) + \frac{\delta^2 t^2}{2}\frac{n(37-15n+2n^2)}{6},$$

$$t=4: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-7) + \frac{\delta^2 t^2}{2}\frac{n(73-21n+2n^2)}{6}.$$

Example 5.5 (Sums of cubes in central differences).

$$t=0: \quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2} n(1+n) + \tfrac{\delta^2 t^3}{2} \tfrac{n(1+3n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2} \tfrac{n(-1+n+4n^2+2n^3)}{24},$$

$$t=1: \quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2} n(n-1) + \tfrac{\delta^2 t^3}{2} \tfrac{n(1-3n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2} \tfrac{n(1+n-4n^2+2n^3)}{24},$$

$$t=2: \quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2} n(n-3) + \tfrac{\delta^2 t^3}{2} \tfrac{n(13-9n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2} \tfrac{n(-21+25n-12n^2+2n^3)}{24},$$

$$t=3: \quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2} n(n-5) + \tfrac{\delta^2 t^3}{2} \tfrac{n(37-15n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2} \tfrac{n(-115+73n-20n^2+2n^3)}{24},$$

$$t=4: \quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2} n(n-7) + \tfrac{\delta^2 t^3}{2} \tfrac{n(73-21n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2} \tfrac{n(-329+145n-28n^2+2n^3)}{24}.$$

6. Remark about central factorial numbers of the second kind

7. Generalized central factorial numbers of the second kind

Definition 7.1 (Generalized Central factorial numbers of the second kind). *For integers $0 \leq k \leq m$, and an arbitrary integer t,*

$$T_t(m,k) = \frac{\delta^k t^m}{k!},$$

defined by central Newton's formula for powers, evaluated at arbitrary integer t

$$n^m = \sum_{k=0}^{\infty} T_t(m,k)(n-t)^{[k]}.$$

Proposition 7.2 (Generalized Central factorial sums). *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^{m} \left[\binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1} \right] k! T_t(m,k)$$

8. Double and triple sums of powers in terms of central differences

Proposition 8.1 (Double sums of powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^2 n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg[+\binom{n-t+\frac{k}{2}+2}{k+2} - \binom{2-t+\frac{k}{2}}{k+2}\Sigma^0 n^0 - \binom{1-t+\frac{k}{2}}{k+1}\Sigma^1 n^0 + \binom{n-t+\frac{k}{2}+1}{k+2} - \binom{1-t+\frac{k}{2}}{k+2}\Sigma^0 n^0 - \binom{0-t+\frac{k}{2}}{k+1}\Sigma^1 n^0\Bigg]\delta^k t^m$$

Proposition 8.2 (Triple sums of powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^3 n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg[+ \binom{n-t+\frac{k}{2}+3}{k+3} - \binom{3-t+\frac{k}{2}}{k+3}\Sigma^0 n^0 - \binom{2-t+\frac{k}{2}}{k+2}\Sigma^1 n^0 - \binom{1-t+\frac{k}{2}}{k+1}\Sigma^2 n^0 + \binom{n-t+\frac{k}{2}+2}{k+3} - \binom{2-t+\frac{k}{2}}{k+3}\Sigma^0 n^0 - \binom{1-t+\frac{k}{2}}{k+2}\Sigma^1 n^0 - \binom{0-t+\frac{k}{2}}{k+1}\Sigma^2 n^0\Bigg]\delta^k t^m$$

Example 8.3 (Double sum of squares in central differences).

$$t=0: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\, n(1+n) + \tfrac{\delta t^2}{2}\, \tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\, \tfrac{n(1+n)^2(2+n)}{12},$$

$$t=1: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\, n(1+n) + \tfrac{\delta t^2}{2}\, \tfrac{n(-1+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\, \tfrac{n^2(-1+n^2)}{12},$$

$$t=2: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\, n(1+n) + \tfrac{\delta t^2}{2}\, \tfrac{n(-4-3n+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\, \tfrac{n(10+5n-4n^2+n^3)}{12},$$

$$t=3: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\, n(1+n) + \tfrac{\delta t^2}{2}\, \tfrac{n(-7-6n+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\, \tfrac{n(32+23n-8n^2+n^3)}{12},$$

$$t=4: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\, n(1+n) + \tfrac{\delta t^2}{2}\, \tfrac{n(-10-9n+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\, \tfrac{n(66+53n-12n^2+n^3)}{12}.$$

Example 8.4 (Triple sum of squares in central differences)**.**

$$t=0: \quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(6+11n+6n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(18+45n+40n^2+15n^3+2n^4)}{120},$$

$$t=1: \quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-2-n+2n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(-2-5n+5n^3+2n^4)}{120},$$

$$t=2: \quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-10-13n-2n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(58+65n-5n^3+2n^4)}{120},$$

$$t=3: \quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-18-25n-6n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(198+255n+40n^2-15n^3+2n^4)}{120},$$

$$t=4: \quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-26-37n-10n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(418+565n+120n^2-25n^3+2n^4)}{120}.$$

9. Multifold sums of powers in terms of central differences

Proposition 9.1 (Multifold sums of powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\begin{aligned}\Sigma^r n^m = \frac{1}{2}\sum_{k=0}^{m} \Bigg\{ & \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \\ & - \sum_{s=0}^{r-1} \left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \Sigma^s n^0 \Bigg\} \delta^k t^m\end{aligned}$$

Example 9.2 (Examples of multifold binomial sums)**.**

$$t=0: \quad \Sigma^r\, n^m = \tfrac{1}{2}\sum_{k=0}^{m}\Bigg\{\tbinom{n+\frac{k}{2}+r}{k+r}+\tbinom{n+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\Big[\tbinom{r-s+\frac{k}{2}}{k+r-s}+\tbinom{r-1-s+\frac{k}{2}}{k+r-s}\Big]\tbinom{s+n-1}{s}\Bigg\}\delta^k 0^m,$$

$$t=1: \quad \Sigma^r\, n^m = \tfrac{1}{2}\sum_{k=0}^{m}\Bigg\{\tbinom{n-1+\frac{k}{2}+r}{k+r}+\tbinom{n-1+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\Big[\tbinom{r-s-1+\frac{k}{2}}{k+r-s}+\tbinom{r-2-s+\frac{k}{2}}{k+r-s}\Big]\tbinom{s+n-1}{s}\Bigg\}\delta^k 1^m,$$

$$t=2: \quad \Sigma^r\, n^m = \tfrac{1}{2}\sum_{k=0}^{m}\Bigg\{\tbinom{n-2+\frac{k}{2}+r}{k+r}+\tbinom{n-2+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\Big[\tbinom{r-s-2+\frac{k}{2}}{k+r-s}+\tbinom{r-3-s+\frac{k}{2}}{k+r-s}\Big]\tbinom{s+n-1}{s}\Bigg\}\delta^k 2^m.$$

10. Multifold sums of powers in generalized central factorial numbers

Proposition 10.1 (Multifold central factorial sums of powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^r\, n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg\{\binom{n-t+\frac{k}{2}+r}{k+r}+\binom{n-t+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\Bigg[\binom{r-s-t+\frac{k}{2}}{k+r-s}+\binom{r-1-s-t+\frac{k}{2}}{k+r-s}\Bigg]\binom{s+n-1}{s}\Bigg\}k!T_t(m,k)$$

t	$T_t(2m,2k)$	OEIS
0	$1,0,1,0,1,1,0,1,5,1,0,1,21,14,1,0,1,85,147,30,1,\cdots$	A269945
1	$1,1,1,1,7,1,1,31,20,1,1,127,231,42,1,1,511,\cdots$	A394692
2	$1,4,1,16,25,1,64,301,65,1,256,3025,1701,126,1,\cdots$	A395456
3	$1,9,1,81,55,1,729,1351,140,1,6561,26335,6951,266,1,\cdots$	A395457

Example 10.2 (Sums of squares in central factorial numbers).

$$t=0:\quad \Sigma^1 n^2 = n\cdot 0 + \tfrac{1}{2}n(1+n)\cdot 0 + \tfrac{1}{6}n(1+3n+2n^2)\cdot 1,$$

$$t=1:\quad \Sigma^1 n^2 = n\cdot 1 + \tfrac{1}{2}n(n-1)\cdot 2 + \tfrac{1}{6}n(1-3n+2n^2)\cdot 1,$$

$$t=2:\quad \Sigma^1 n^2 = n\cdot 4 + \tfrac{1}{2}n(n-3)\cdot 4 + \tfrac{1}{6}n(13-9n+2n^2)\cdot 1,$$

$$t=3:\quad \Sigma^1 n^2 = n\cdot 9 + \tfrac{1}{2}n(n-5)\cdot 6 + \tfrac{1}{6}n(37-15n+2n^2)\cdot 1,$$

$$t=4:\quad \Sigma^1 n^2 = n\cdot 16 + \tfrac{1}{2}n(n-7)\cdot 8 + \tfrac{1}{6}n(73-21n+2n^2)\cdot 1,$$

$$t=5:\quad \Sigma^1 n^2 = n\cdot 25 + \tfrac{1}{2}n(n-9)\cdot 10 + \tfrac{1}{6}n(121-27n+2n^2)\cdot 1.$$

Example 10.3 (Sums of cubes in central factorial numbers).

$$\begin{aligned} t=0:\quad \Sigma^1 n^4 &= n\cdot 0 + \tfrac{1}{2}n(1+n)\cdot 0 + \tfrac{1}{6}n(1+3n+2n^2)\cdot 1 \\ &\quad + \tfrac{1}{8}n(-1+n+4n^2+2n^3)\cdot 0 + \tfrac{1}{10}n(-2-5n+5n^3+2n^4)\cdot 1, \end{aligned}$$

$$\begin{aligned} t=1:\quad \Sigma^1 n^4 &= n\cdot 1 + \tfrac{1}{2}n(n-1)\cdot 5 + \tfrac{1}{6}n(1-3n+2n^2)\cdot 7 \\ &\quad + \tfrac{1}{8}n(1+n-4n^2+2n^3)\cdot 4 + \tfrac{1}{10}n(-2+5n-5n^3+2n^4)\cdot 1, \end{aligned}$$

$$\begin{aligned} t=2:\quad \Sigma^1 n^4 &= n\cdot 16 + \tfrac{1}{2}n(n-3)\cdot 34 + \tfrac{1}{6}n(13-9n+2n^2)\cdot 25 \\ &\quad + \tfrac{1}{8}n(-21+25n-12n^2+2n^3)\cdot 8 + \tfrac{1}{10}n(18-45n+40n^2-15n^3+2n^4)\cdot 1, \end{aligned}$$

$$\begin{aligned} t=3:\quad \Sigma^1 n^4 &= n\cdot 81 + \tfrac{1}{2}n(n-5)\cdot 111 + \tfrac{1}{6}n(37-15n+2n^2)\cdot 55 \\ &\quad + \tfrac{1}{8}n(-115+73n-20n^2+2n^3)\cdot 12 + \tfrac{1}{10}n(298-275n+120n^2-25n^3+2n^4)\cdot 1, \end{aligned}$$

$$\begin{aligned} t=4:\quad \Sigma^1 n^4 &= n\cdot 256 + \tfrac{1}{2}n(n-7)\cdot 260 + \tfrac{1}{6}n(73-21n+2n^2)\cdot 97 \\ &\quad + \tfrac{1}{8}n(-329+145n-28n^2+2n^3)\cdot 16 + \tfrac{1}{10}n(1318-805n+240n^2-35n^3+2n^4)\cdot 1, \end{aligned}$$

$$\begin{aligned} t=5:\quad \Sigma^1 n^4 &= n\cdot 625 + \tfrac{1}{2}n(n-9)\cdot 505 + \tfrac{1}{6}n(121-27n+2n^2)\cdot 151 \\ &\quad + \tfrac{1}{8}n(-711+241n-36n^2+2n^3)\cdot 20 + \tfrac{1}{10}n(3798-1755n+400n^2-45n^3+2n^4)\cdot 1. \end{aligned}$$

11. Multifold sums of powers in binomial form

Lemma 11.1 (Multifold sum of zero powers)**.** *For integers $r \geq 0$ and $n \geq 1$*

$$\Sigma^r n^0 = \binom{r+n-1}{r}$$

Proof. (1) Let $r = 0$, then we have $\Sigma^0 n^0 = 1$, by definition (3.1).

(2) Let $r = 1$, then we have $\Sigma^1 n^0 = \sum_{k=1}^{n} 1 = \binom{n}{1}$.

(3) Let $r = 2$, then we have $\Sigma^2 n^0 = \sum_{k=1}^{n} \binom{k}{1} = \binom{n+1}{2}$.

(4) Let $r = 3$, then we have $\Sigma^3 n^0 = \sum_{k=1}^{n} \binom{k+1}{2} = \binom{n+2}{3}$.

(5) Similarly, for arbitrary r, by hockey stick identity $\sum_{k=1}^{n} \binom{k}{r} = \binom{n+1}{r+1}$, thus, the claim follows $\Sigma^r n^0 = \sum_{k=1}^{n} \Sigma^{r-1} k^0 = \sum_{k=1}^{n} \binom{k+r-2}{r-1} = \binom{r+n-1}{r}$.

□

Proposition 11.2 (Multifold binomial sums of powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^r n^m = \frac{1}{2} \sum_{k=0}^{m} \Bigg\{ \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1} \left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \Bigg\} \delta^k t^m$$

12. Related works

13. Framework for sums of powers

14. Conclusions

15. Mathematica programs

Use this GitHub repository to validate the results using Mathematica programs.

Mathematica Function	**Validates / Prints**
`MultifoldSumOfPowersRecurrence[r, n, m]`	Computes $\Sigma^r n^m$, (3.1)
`CentralFactorial[n, k]`	Computes $n^{[k]}$, (2.2)
`ValidateHalvedCentralBinomials[20]`	Validates Lem. (4.2)
`ValidateNewtonsFormulaForPowers[10]`	Validates Prop. (4.3)
`ValidateHalvedCentralFactorialPowers[5]`	Validates Prop. (4.4)
`ValidateCenteredHockeyStickIdentity[5]`	Validates Lem. (4.6)
`ValidateOrdinarySumsOfPowersDecomposition[10]`	Validates Prop. (5.2)
`ValidateDoubleSumsOfPowersDecomposition[5]`	Validates Prop. (8.1)
`ValidateTripleSumsOfPowersDecomposition[5]`	Validates Prop. (8.2)
`ValidateMultifoldSumsOfPowers[3]`	Validates Prop. (9.1)